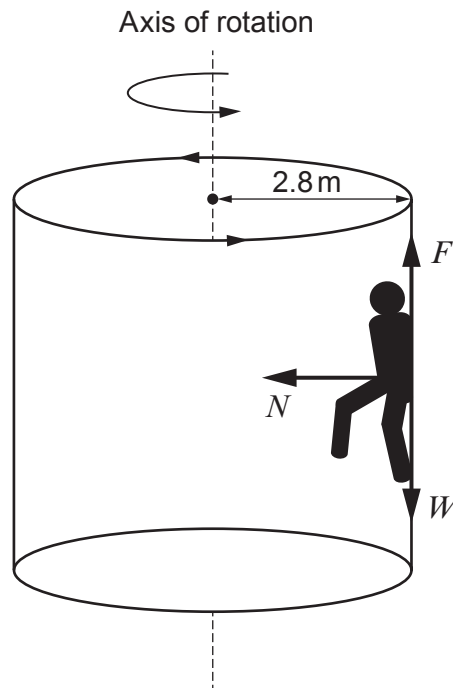


2. A fairground ride consists of a large empty cylinder of radius 2.80 m with its axis of rotation vertical. A person of mass 66.2 kg walks into the cylinder and stands against its inner wall. The cylinder is rotated around the axis. When the rotation rate is 36.0 revolutions per minute the floor is lowered, but the person remains fixed in place against the wall.

This is a student's diagram.



- (a) Define the *radian*.

[1]

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- (b) Calculate:

- (i) the period of rotation, T ;

[2]

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- (ii) the speed, v , at which the person is moving in the circle.

[2]

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- (c) Three forces act on the person: the normal contact force from the wall, N , the weight, W , of the person, and the frictional force, F , from the wall. These are shown in the student's diagram.

Explain why the values of N and F are approximately 2600 N and 650 N respectively.

[4]

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- (d) The maximum value that a frictional force can take is μN where μ is a dimensionless constant.

- (i) Show that the person does not slide down the wall provided that μ is larger than approximately 0.25. [2]

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- (ii) A student says that as the cylinder slows down the floor will need to be raised to support the person when the angular velocity reduces below a certain value. Justify this and determine this angular velocity if μ has a value of 0.45. [4]

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